OĞUZHAN KURNAZ

PhD | Assistant Professor, Bursa Technical University

Spoofing-Aware Speaker Verification · Speech Processing · Machine & Deep Learning

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PROFILE

Assistant Professor at Bursa Technical University with a PhD in Electrical-Electronics Engineering, specializing in **spoofing-aware speaker verification (SASV)**, speech and audio processing, and machine/deep learning. Former **TÜBİTAK 2214-A visiting researcher at the University of Eastern Finland**, with publications in *IEEE Signal Processing Letters*, *IEEE Transactions on Audio, Speech and Language Processing*, *ICASSP* and *Interspeech*, and active participation in the ASVspoof 5, NIST SRE 2024 and WildSpoof challenges. Brings a solid industrial background in mechatronics, PLC/robot programming and automation.

14	**3**	**2214-A**	**1001**
Published & accepted papers	International challenges: ASVspoof 5 · NIST SRE24 · WildSpoof	TÜBİTAK international research fellow (UEF, Finland)	Researcher in a TÜBİTAK 1001 research project

EDUCATION

PhD, Electrical-Electronics Engineering — **2019 – 2025**
Bursa Technical University
Thesis: *A Decision Cost-Based Learning Approach for Spoofing-Aware Speaker Verification*

Visiting PhD Researcher, School of Computing — **2023 – 2024**
University of Eastern Finland, Joensuu, Finland
TÜBİTAK 2214-A International Research Fellowship Programme for PhD Students

MSc, Electrical-Electronics Engineering — **2016 – 2019**
Erzurum Technical University
Thesis: *Image-Based Control of Linear Robotic System*

BSc, Mechatronics Engineering — **2011 – 2016**
Erciyes University, Kayseri
Focus areas: Robotics, PLC, Automation

WORK EXPERIENCE

Assistant Professor — **08/2026 – Present**
Bursa Technical University

- Leads research on **spoofing-aware speaker verification**, speech processing and machine/deep learning; researcher in the **TÜBİTAK 1001** project on deep and self-supervised learning for SASV.
- Teaches courses in Python, C, object-oriented programming and machine learning.

Research Assistant — **06/2018 – 08/2026**
Bursa Technical University

- Conducted research on **spoofing-aware speaker verification**, developing novel machine learning and deep learning methods to make speaker verification systems robust against spoofing attacks.
- Assisted courses in Python, C, object-oriented programming and machine learning.

Visiting Researcher — **12/2023 – 12/2024**
University of Eastern Finland, School of Computing

- One-year research stay funded by the **TÜBİTAK 2214-A** International Research Fellowship Programme for PhD Students.
- Built international collaborations in speech processing, leading to participation in the **ASVspoof 5** and **NIST SRE 2024** challenges.
- Delivered a course on spoofing-aware speaker verification at a university-organized summer school.

Mechatronics Engineer
10/2017 – 06/2018

Birleşim Electric, Erzurum

- Monitored and managed electrical panel manufacturing; troubleshooting, problem-solving and documentation of production issues.
- Designed electrical panel connections using EPLAN.

Mechatronics Engineer
10/2015 – 09/2016

ELMOS Automation, Kayseri

- Managed production of electrical panels and robot-based automation systems; troubleshooting and technical reporting.
- Programmed industrial robots and Programmable Logic Controllers (PLCs); designed panel connections in EPLAN.

■ RESEARCH PROJECTS & THESES

TÜBİTAK 1001 – Researcher
Accepted 19.01.2026

Deep Learning and Self-Supervised Learning Solutions for Spoofing-Aware Speaker Verification

Jointly addresses automatic speaker verification (ASV) and countermeasure (CM) tasks against synthetic speech and replay attacks. Investigates information fusion at both score and deep-feature level (non-linear score fusion, attention-based feature fusion), the effect of data augmentation, and the direct adaptation of self-supervised learning (SSL) representations and adapters within a unified SASV framework on the ASVspoof 5 dataset. Team: one PI, two researchers and two PhD scholars across six work packages.

TÜBİTAK 2214-A Research Project
2023 – 2024

Investigation of Integration Methods for Spoofing Attack Detection and Speaker Verification Systems in a Spoofing-Aware Speaker Verification Framework (University of Eastern Finland)

Explores the fusion of speaker verification and spoofing detection systems at both embedding and score levels to build a unified spoofing-aware speaker verification system; investigates alternative evaluation metrics and benchmarks the proposed methods on ASVspoof 2019.

PhD Thesis
2019 – 2025

A Decision Cost-Based Learning Approach for Spoofing-Aware Speaker Verification (BTU)

Proposes SASV methods integrating state-of-the-art models (**ECAPA-TDNN, WavLM, AASIST**) through score- and embedding-level fusion and directly optimizes the architecture-agnostic detection cost function (**a-DCF**). Experiments on ASVspoof 2019 and ASVspoof 5 show significant gains in separating target, non-target and spoof trials, contributing to more secure biometric authentication.

Master's Thesis
2015 – 2016

Image-Based Control of Linear Robotic System (ETU)

Established the foundational framework of a **Robotic Mushroom Harvesting System** integrating a PLC, a vision-based camera module and motor-driven actuators: the camera detects and localizes the mushroom, and the PLC coordinates the motors to reach and harvest it autonomously.

■ PUBLICATIONS

Journal Articles

[1] **O. Kurnaz**, J. Mishra, T. H. Kinnunen and C. Hanilçi, "Joint Optimization of Speaker and Spoof Detectors for Spoofing-Robust Automatic Speaker Verification," *IEEE Transactions on Audio, Speech and Language Processing*, 2026, doi: 10.1109/TASLPRO.2026.3688932.

[2] **O. Kurnaz**, J. Mishra, T. H. Kinnunen and C. Hanilçi, "Optimizing a-DCF for Spoofing-Robust Speaker Verification," *IEEE Signal Processing Letters*, vol. 32, pp. 1081–1085, 2025, doi: 10.1109/LSP.2025.3545290.

[3] K. K. Ayten and **O. Kurnaz**, "Real-Time Implementation of Image Based PLC Control for a Robotic Platform," *Balkan Journal of Electrical and Computer Engineering*, vol. 7, no. 1, pp. 64–71, 2019, doi: 10.17694/bajece.487212.

Conference Papers

[4] **O. Kurnaz**, J. Mishra, T. Kinnunen and C. Hanilçi, "Joint Optimization of ASV and CM Tasks: Btuef Team's Submission for WildSpoof Challenge," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2026)*, Barcelona, Spain, 2026, pp. 21910–21912, doi: 10.1109/ICASSP55912.2026.11461784.

[5] D. Colibro, C. Vair, Y. Tu, J. Li, Z. Huang, Y. Chen, K. A. Lee, M.-W. Mak, J. Mishra, V. Singh, X. Xuan, M. Chhibber, **O. Kurnaz**, T. Kinnunen, S. Lee, C. Jung, K. Nam, J. S. Chung and S. Wang, "I4U's Official and Streamlined Audio Systems for NIST SRE24," in *Proc. Odyssey 2026*, pp. 191–198, doi: 10.21437/Odyssey.2026-28.

[6] **O. Kurnaz**, "Calibration-Aware Joint Optimization for Spoofing-Robust Speaker Verification via Auxiliary C_{llr} Losses," in *Proc. 6th International Conference on Advanced Engineering Technologies (ICADET 2026)* (accepted).

[7] **O. Kurnaz**, "QMF-SASV: Quality-Measure Fusion Using Duration and Embedding-Norm Cues for Spoofing-Aware Speaker Verification," in *Proc. 6th International Conference on Advanced Engineering Technologies (ICADET 2026)* (accepted).

[8] A. Buker, **O. Kurnaz**, S. Bekiryazici, S. C. Demirtas and C. Hanilci, "Evaluating Parameter Sharing for Spoofing-Aware Speaker Verification: A Case Study on the ASVspoof 5 Dataset," in *Proc. Interspeech 2025*.

[9] **O. Kurnaz**, T. Kinnunen and C. Hanilci, "Investigating the Potential of Multi-Stage Score Fusion in Spoofing-Aware Speaker Verification," *33rd Signal Processing and Communications Applications Conference (SIU 2025)*, İstanbul, Türkiye.

[10] **O. Kurnaz**, S. C. Demirtaş, J. Mishra, A. Buker and C. Hanilçi, "Spoofing-Robust Speaker Verification Using Parallel Embedding Fusion: BTU Speech Group's Approach for ASVspoof5 Challenge," in *Proc. The Automatic Speaker Verification Spoofing Countermeasures Workshop (ASVspoof 2024)*, pp. 138–143, doi: 10.21437/ASVspoof.2024-20.

[11] **O. Kurnaz** and C. Hanilçi, "Multi-image Crowd Counting Using Multi-column Convolutional Neural Network," in *Proc. Sixth International Congress on Information and Communication Technology*, Lecture Notes in Networks and Systems, vol. 236, Springer, Singapore, 2022, doi: 10.1007/978-981-16-2380-6_20.

[12] G. Aydemir, **O. Kurnaz**, T. Bekiryazıcı, A. Avcı and M. Kocakulak, "Driver Drowsiness Detection using MobileNets and Long Short-term Memory," *2021 13th International Conference on Electrical and Electronics Engineering (ELECO)*, Bursa, Turkey, 2021, pp. 220–223, doi: 10.23919/ELECO54474.2021.9677724.

[13] **O. Kurnaz** and C. Hanilçi, "Multi-Image Crowd Density Estimation using Multi Column Deep Neural Network," *2020 28th Signal Processing and Communications Applications Conference (SIU)*, Gaziantep, Turkey, 2020, pp. 1–4, doi: 10.1109/SIU49456.2020.9302347.

[14] **O. Kurnaz** and C. Hanilçi, "Features and Regression Techniques for Crowd Density Estimation: A Comparison," *2019 11th International Conference on Electrical and Electronics Engineering (ELECO)*, Bursa, Turkey, 2019, pp. 1075–1079, doi: 10.23919/ELECO47770.2019.8990540.

Under Review

[15] **O. Kurnaz**, "Adaptive Multi-Operating-Point Optimization for Spoofing-Aware Speaker Verification with a Common Decision Threshold," submitted to *Journal of Innovative Science and Engineering*, 2026.

■ CERTIFICATES & COURSES

Artificial Intelligence and Machine Learning *Certificate of Achievement (module exam passed)*	YÖK Data Analysis School – Marmara University, in collaboration with METU, ITU and Boğaziçi University	**2025 – 2026**
Artificial Intelligence and Facilitating Tools *Certificate of Achievement (module exam passed)*	YÖK Data Analysis School – Marmara University, in collaboration with METU, ITU and Boğaziçi University	**2025 – 2026**
Statistical Machine Learning	University of Eastern Finland	**2024**
Scientific Presentation	University of Eastern Finland	**2024**
Deep Learning and Applications	Udemy	**2019**
Machine Learning based Python	Udemy	**2019**

■ SKILLS, TEACHING & LANGUAGES

Areas of Expertise: Machine & Deep Learning for Image and Signal Processing · Speaker Verification and Anti-Spoofing · Speech and Audio Processing

Programming Languages: Python, C#, C (Advanced)

Engineering & Scientific Tools: MATLAB, LabVIEW, TIA Portal (Intermediate); EPLAN

Industrial Automation: PLC programming, industrial robot programming, electrical panel design

Office Applications: Microsoft Office (Advanced)

Languages: English – B2 (advanced proficiency in reading, writing and speaking) · Russian – Beginner (A1)

Teaching: Python Programming · C Programming · Object-Oriented Programming · Machine Learning with Python · Summer-school course on Spoofing-Aware Speaker Verification (UEF)

■ REFERENCES

Prof. Cemal HANİLÇİ	Bursa Technical University	Professor	cemal.hanilci@btu.edu.tr
Prof. Hakan GÜRKAN	Bursa Technical University	Professor	hakan.gurkan@btu.edu.tr
Prof. Tomi KINNUNEN	University of Eastern Finland	Professor	tomi.kinnunen@uef.fi